Pointwise Einstein approximation on the Miyaoka–Yau numerical boundary

24 September 2026. This continuation removes externally prescribed saturation and topological bubble-exclusion hypotheses at a realizable numerical boundary. The Chern–Weil identities are classical; the new project conclusion is their composition with compactness and actual-form supply. Global priority is not asserted.

1. The realization statement

Put $T_\pm = 2\chi \pm 3\tau$ and

$$\delta(g) = \| \operatorname{Ric}_g^\sharp + I \|_{\infty,\mathrm{op}}.$$

Use tensor Hilbert–Schmidt norms for Ricci, exterior norms for two-forms, and endomorphism Hilbert–Schmidt norms for $W^\pm$.

Theorem A. For every $D > 0$ there exists $\varepsilon(D) > 0$ with the following property. Let M be a closed connected oriented smooth four-manifold with

$$A := T_+ = 3T_- > 0. \tag{1}$$

Let g have diameter at most D. Suppose a Spin^c structure is actually monopole-solvable for every metric conformally related to g, and its first Chern class a satisfies

$$p = (a_g^+)^2 \geqslant A. \tag{2}$$

If $\delta(g) < \varepsilon(D)$, the **same oriented smooth manifold** admits a complex-hyperbolic metric h with $\operatorname{Ric}_h = -h$. Its universal Riemannian cover is the complex hyperbolic plane of holomorphic sectional curvature $-2/3$.

No input Kähler structure, π_1 or π_2 hypothesis, Weyl bound, class-square cost, Ricci-derivative bound, or functional-saturation assumption is imposed. The class may depend on g, but is fixed throughout its conformal class. The threshold is uniform over the stated manifolds and classes.

Condition (1) is exactly $\chi = 3\tau > 0$, not a general condition on four-manifolds. Complex-hyperbolic existence fully supplies the Einstein output required by the adopted Yau26 question on these inputs. The remaining limitations are the diameter and numerical/gauge **input** conditions.

2. Saturation follows from the numerical boundary

Write

$$S = \int s^2, \quad P = \int |\operatorname{Ric}^\circ|^2, \quad W_\pm = \int |W^\pm|^2, \quad V = \operatorname{Vol}(g),$$

$$\mathfrak{D}_2 = \frac{3}{2}(S + 12W_+) - 72\pi^2 p, \qquad q = \mathfrak{D}_2/V.$$

The actual-monopole inequalities give

$$S \geqslant 32\pi^2 p \geqslant 32\pi^2 A, \qquad \mathfrak{D}_2 \geqslant 0. \tag{3}$$

The opposite-orientation Chern–Weil formula is exactly

$$S - 32\pi^2 A + 48W_- = 12P. \tag{4}$$

Consequently

$$0 \leqslant W_- \leqslant P/4, \qquad 0 \leqslant S - 32\pi^2 A \leqslant 12P.$$

Eliminating W_+ with the other Chern–Weil formula first gives

$$\mathfrak{D}_2 = \frac{9}{8}S + \frac{9}{2}P + 36\pi^2 A - 72\pi^2 p.$$

Substitution of (4) gives a second exact balance:

$$\boxed{\mathfrak{D}_2 + 72\pi^2(p - A) + 54W_- = 18P.} \tag{5}$$

All terms on its left are nonnegative. Orthogonal trace decomposition and $|\mathrm{Ric} + g|^2 \leqslant 4\delta^2$ now imply

$$\boxed{q \leqslant 72\delta^2, \qquad W_-/V \leqslant \delta^2.} \tag{6}$$

Thus the previously independent gauge deficit and opposite Weyl energy are forced small by the **original pointwise input**. This implication is specific to (1)–(2), not a general property of almost-Einstein metrics.

Volume is also controlled:

$$V \geqslant \frac{2\pi^2 A}{(1+\delta)^2}, \qquad |V - 2\pi^2 A| \leqslant (2\delta + 4\delta^2)V. \tag{7}$$

The second bound follows from $|S/V - 16| \leqslant 32\delta + 16\delta^2$ and $0 \leqslant S - 32\pi^2 A \leqslant 48\delta^2 V$. No volume normalization is assumed.

3. Both Weyl halves exclude concentration without topology

Suppose Theorem A were false for a fixed D. Choose counterexamples (M_j, g_j, a_j) with $\delta_j \to 0$ and no complex-hyperbolic metric on their oriented smooth types. For large j, $\delta_j \leqslant 1/2$. The positive integer $A_j = T_+(M_j)$ is at least one, so (7) gives $V_j \geqslant 8\pi^2/9$. Ricci is bounded on both sides. Bishop comparison and the diameter bound give $V_j \leqslant V_0(D)$ and

$$c(D)r^4 \leqslant \mathrm{Vol}_{g_j} B_r(x) \leqslant Cr^4 \quad (0 < r \leqslant 1). \tag{8}$$

Set $r_j = \fint |\mathrm{Ric} + g_j|^2$ and $\kappa_j = r_j + q_j \leqslant 76\delta_j^2$. The elementary amplitude consequence of the classical completed-square estimate, recorded in Section 4 of the gauge-realization proof, gives

$$\int_E |W^+|^2 \leqslant \frac{4}{3}\,\mathrm{Vol}(E) + \frac{2}{3}\kappa_j V_j \tag{9}$$

for every measurable set E. Meanwhile (6) gives

$$\int_{M_j} |W^-|^2 \to 0. \tag{10}$$

Both halves therefore vanish on every shrinking ball in a harmonic-radius blowup. There is no half-flat ALE core left to exclude by topology.

For clarity, the compactness argument is the following. If the minimum $C^{1,\alpha}$ harmonic radius $\rho_j \to 0$, rescale by ρ_j^{-2}. All harmonic radii are at least one, the base radius is one, and Ricci tends uniformly

to zero. Harmonic-coordinate compactness, (8), and the metric equation give a complete smooth Ricci-flat limit with Euclidean lower volume growth. Subtracting the limiting metric equation improves convergence to strong local $W^{2,p}$, all finite p, recovering the Hölder exponent used to normalize the harmonic radius. This uses no derivative of Ricci. Section 4 of compact realization supplies this analytic argument before its ALE step.

Scale invariance of Weyl energy, (8)–(10), and strong curvature convergence give $W^+ = W^- = 0$ on the limit. Thus it is flat. A complete smooth flat manifold with Euclidean lower volume growth at all scales is Euclidean: an infinite deck group lowers the growth order, and a finite Euclidean isometry group cannot act freely unless trivial. Harmonic charts on a larger Euclidean ball transfer back and contradict the normalized base harmonic radius. Hence harmonic radii are uniformly positive.

This particular proof needs neither a Jiang–Naber total-energy bound, nor ALE asymptotics or classification, nor any homotopy restriction. Global diameter and (8) give a finite atlas. After diffeomorphisms and a subsequence the metrics converge on a common smooth oriented manifold in $C^{1,\alpha}$ and strongly in finite $W^{2,p}$ to a smooth metric h with $\mathrm{Ric}_h = -h$ and $W_h^- = 0$. No convergence of every derivative of the approximate metrics is asserted.

4. Global complex structure and the exact output

The retained actual-form theorem supplies global self-dual forms ω_j, of uniformly bounded amplitude, with

$$\fint (|\omega_j|^2 - 2)^2 \to 0, \qquad \fint |\nabla \omega_j|^2 \to 0. \tag{11}$$

These first-order estimates require no negative-square budget. Transfer the forms to the common smooth manifold. The volume bounds and C^1 metric convergence give a fixed-metric H^1 bound. Rellich compactness gives strong L^2 convergence to ω, and (11) gives $\nabla_h \omega = 0$, $|\omega|^2 = 2$. Thus ω is smooth and defines a global parallel orthogonal complex structure J. It is integrable and has the prescribed orientation; no double cover occurs.

A Kähler–Einstein surface with $W^- = 0$ has vanishing Bochner curvature, hence constant holomorphic sectional curvature. With $\mathrm{Ric} = -h$, that curvature is $-2/3$. Completeness identifies the universal cover with the complex hyperbolic plane. Pull this exact metric back to each sufficiently late M_j, contradicting the counterexample choice. This proves the uniform $\varepsilon(D)$.

5. Suppliers, examples, and the actual increment

An actual conformally solvable class with $a^2 \geqslant T_+ > 0$ automatically satisfies (2), because $p = a^2 + \|a_g^-\|^2$. A nonzero ordinary SW invariant in nonnegative formal dimension, with applicable chamber conditions, is a classical supplier. A formal cohomology vector is not. For $b_+ = 1$, retain the metricwise polarized/chamber argument.

Compact complex-hyperbolic surfaces and their canonical classes are real equality models:

$$S = 32\pi^2 A, \quad P = W_- = \mathfrak{D}_2 = 0, \quad p = A, \quad V = 2\pi^2 A.$$

The theorem does not assume the input smooth type already supports such a structure. Conversely, on a smooth type satisfying the stated gauge supply and (1), but not supporting complex-hyperbolic geometry, it gives a diameter-dependent **pointwise defect gap**, not merely a joint deficit gap.

The zero-defect classification is classical Seiberg–Witten/Miyaoka–Yau rigidity; (4)–(5) are classical identities. The project increment is the uniform approximate-to-exact conclusion, with saturation and earlier topological bubble-exclusion assumptions removed. The older unrestricted integral obstacles remain distinct: they have no diameter hypothesis and cover incompatible numerical regions. Here a compatible numerical boundary produces an actual exact geometry.

No optimal threshold, new non-complex-hyperbolic example, or unrestricted Yau26 solution is claimed.

6. Primary inputs and proof dependencies

- LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Theorem 2.3, Proposition 4.1 and formula (1): actual conformal monopole inequalities and Chern–Weil normalization.
- LeBrun, *Einstein Metrics and Mostow Rigidity*, Math. Res. Lett. 2 (1995), 1–8, DOI: the classical exact rigidity comparison, not a new exact classification.
- Anderson, harmonic-coordinate compactness, as used with the detailed metric-equation argument in the retained proof.

The actual-form construction remains a project PDE dependency. The finite checks below verify only the two exact balances and the equality calibration.